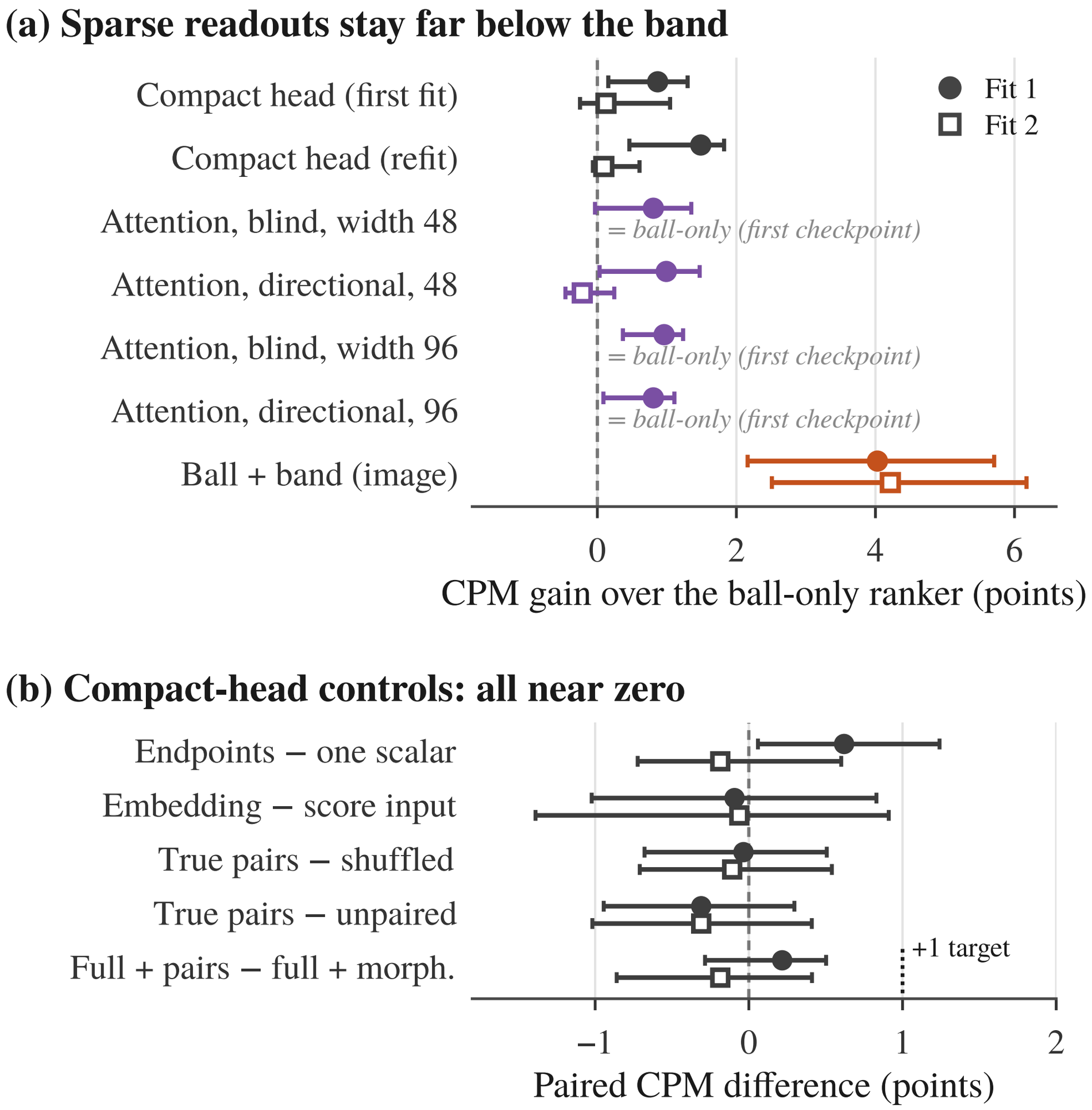
(a) Sparse readouts stay far below the band
Compact head (first fit)
Compact head (refit)
Attention, blind, width 48
Attention, directional, 48
Attention, blind, width 96
Attention, directional, 96
Ball + band (image)
Fit 1
Fit 2
= ball-only (first checkpoint)
= ball-only (first checkpoint)
= ball-only (first checkpoint)
0
2
4
6
CPM gain over the ball-only ranker (points)
(b) Compact-head controls: all near zero
Endpoints − one scalar
Embedding − score input
True pairs − shuffled
True pairs − unpaired
Full + pairs − full + morph.
+1 target
−1
0
1
2
Paired CPM difference (points)